

ADV-DEVOPS EXPERIMENT-2

Aim : To create bucket using S3.

Theory :

An Amazon S3 bucket is a fundamental component within Amazon Web Services (AWS) Simple Storage Service (S3), which is an object storage service. Essentially, an S3 bucket serves as a container for storing objects (files) in the cloud.

In essence, an S3 bucket is a highly scalable, durable, and secure storage solution within AWS, providing a flexible way to store and manage diverse types of data in the cloud.

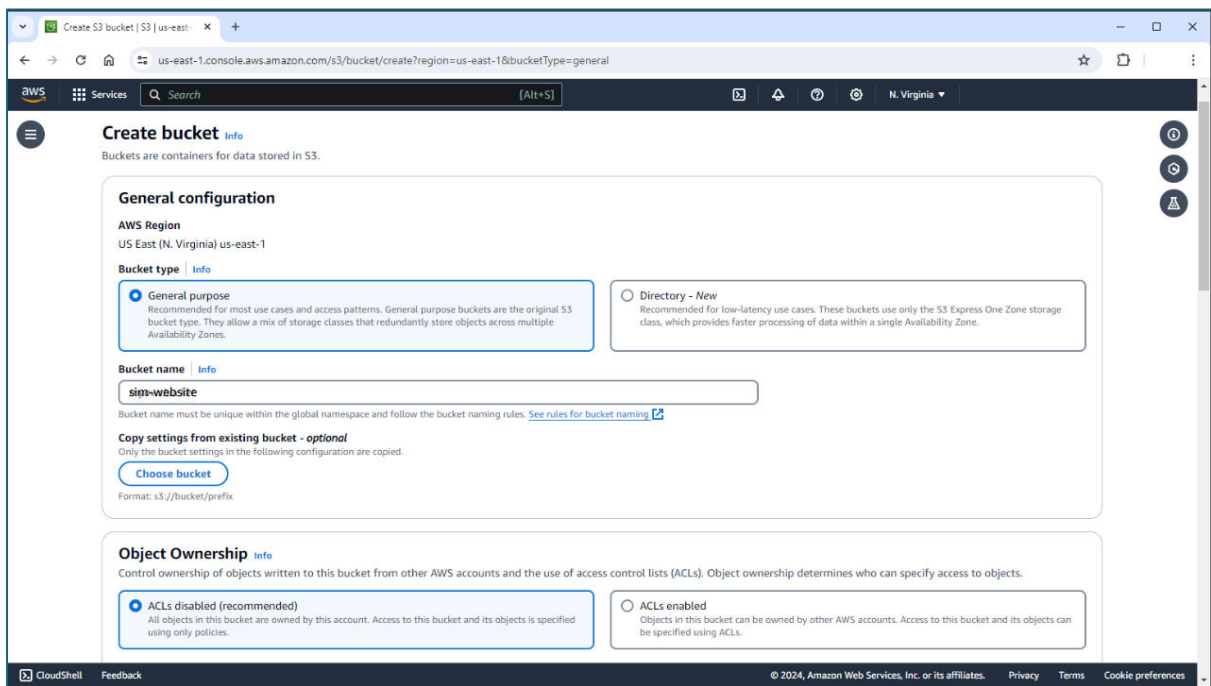
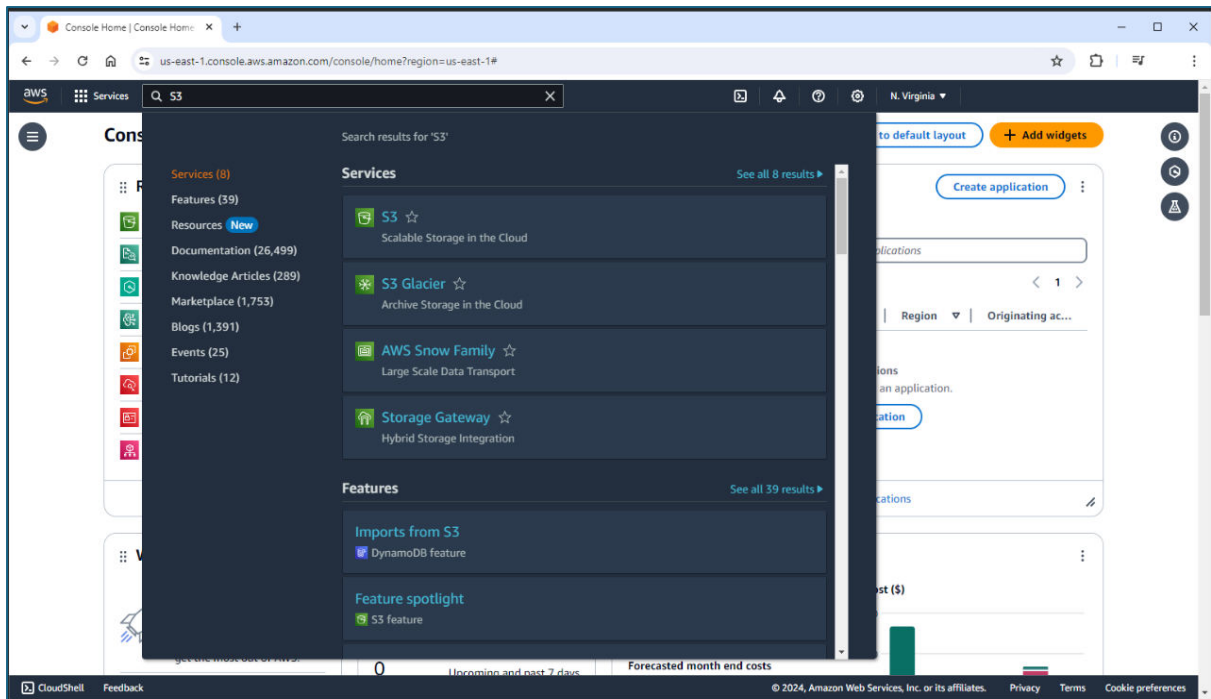
Procedure :

Task 1: Creating a bucket in Amazon S3

1. In the AWS Management Console, choose the search bar and enter S3. Then choose S3 from the search results.
2. Choose Create bucket.
3. For Bucket name, enter sim-website.
4. Choose the scroll bar to scroll down to Object Ownership. For Object Ownership, choose ACLs enabled.
5. Choose the scroll bar to scroll down to Block Public Access settings for this bucket.
6. For Block Public Access settings for this bucket, clear the checkbox for Block all public access. Then, select the box that states I acknowledge that the current settings might result in this bucket and the objects within becoming public.
7. Choose the scroll bar to scroll down to Bucket Versioning.
8. For Bucket Versioning, choose Enable.
9. For Tags, choose Add tag, and enter the following:

Key: Department **Value:** Marketing

10. Choose the scroll bar to scroll down.
11. Choose Create bucket.



Create S3 bucket | S3 | us-east-1

us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1&bucketType=general

aws

Services

Search

[Alt+S]

N. Virginia

Object Ownership

info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

☐ ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☒ ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

⚠

We recommend disabling ACLs, unless you need to control access for each object individually or to have the object writer own the data they upload. Using a bucket policy instead of ACLs to share data with users outside of your account simplifies permissions management and auditing.

Object Ownership

☒ Bucket owner preferred
If new objects written to this bucket specify the bucket-owner-full-control canned ACL, they are owned by the bucket owner. Otherwise, they are owned by the object writer.

☐ Object writer
The object writer remains the object owner.

ℹ

If you want to enforce object ownership for new objects only, your bucket policy must specify that the bucket-owner-full-control canned ACL is required for object uploads. [Learn more](#)

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

ℹ

Block Public Access settings for this account are currently turned on

[Block Public Access settings for this account](#) that are enabled apply even if they are disabled for this bucket.

CloudShell

Feedback

© 2024, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

Create S3 bucket | S3 | us-east-1

us-east-1.console.aws.amazon.com/s3/bucket/create?region=us-east-1&bucketType=general

aws

Services

Search

[Alt+S]

N. Virginia

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

ℹ

Block Public Access settings for this account are currently turned on

[Block Public Access settings for this account](#) that are enabled apply even if they are disabled for this bucket.

☐ Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ Block public access to buckets and objects granted through new access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ Block public access to buckets and objects granted through any access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ Block public access to buckets and objects granted through new public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ Block public and cross-account access to buckets and objects through any public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

⚠

Turning off block all public access might result in this bucket and the objects within becoming public

AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3

CloudShell

Feedback

© 2024, Amazon Web Services, Inc. or its affiliates. [Privacy](#) [Terms](#) [Cookie preferences](#)

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☐ Disable

☒ Enable

Tags - optional (1)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

Key **Value - optional**

Department Marketing [Remove](#)

[Add tag](#)

Default encryption [Info](#)

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the Storage tab of the [Amazon S3 pricing page](#).

Bucket Key

Successfully created bucket "sim-website"

To upload files and folders, or to configure additional bucket settings, choose [View details](#).

[View details](#)

Account snapshot - updated every 24 hours [All AWS Regions](#)

Storage lens provides visibility into storage usage and activity trends. [Learn more](#)

[View Storage Lens dashboard](#)

General purpose buckets **Directory buckets**

General purpose buckets (4) [Info](#) [All AWS Regions](#)

Buckets are containers for data stored in S3.

[Refresh](#) [Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Name	AWS Region	IAM Access Analyzer	Creation date
cloudtrail-awslogs-033742928620-vgaivtv4-isengard-do-not-delete	US East (N. Virginia) us-east-1	View analyzer for us-east-1	March 18, 2021, 14:54:43 (UTC-04:00)
do-not-delete-gatedgarden-audit-033742928620	US West (Oregon) us-west-2	View analyzer for us-west-2	April 1, 2021, 07:00:01 (UTC-04:00)
sagemaker-us-east-1-033742928620	US East (N. Virginia) us-east-1	View analyzer for us-east-1	June 22, 2021, 11:12:16 (UTC-04:00)
sim-website	US East (N. Virginia) us-east-1	View analyzer for us-east-1	June 4, 2024, 15:53:27 (UTC-04:00)

This completes Task 1: Creating a bucket in Amazon S3. Choose Continue.

Task 2: Configuring a static website on Amazon S3

12. In the list of your buckets, choose the name of the bucket that you just created, sim-website.
13. Choose the Properties tab.
14. Choose the scroll bar to scroll to the Static website hosting panel.
15. Choose Edit to the Static website hosting panel, Choose Enable.
16. For Hosting type, keep the default setting Host a static website.

17. Configure the following settings:

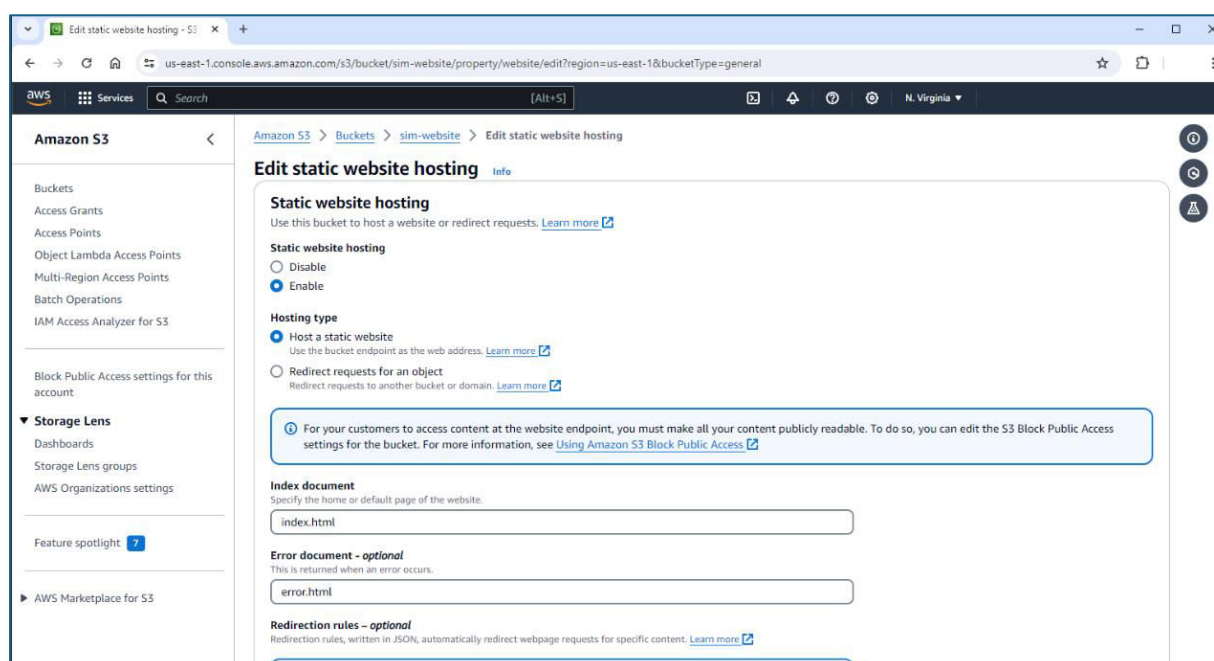
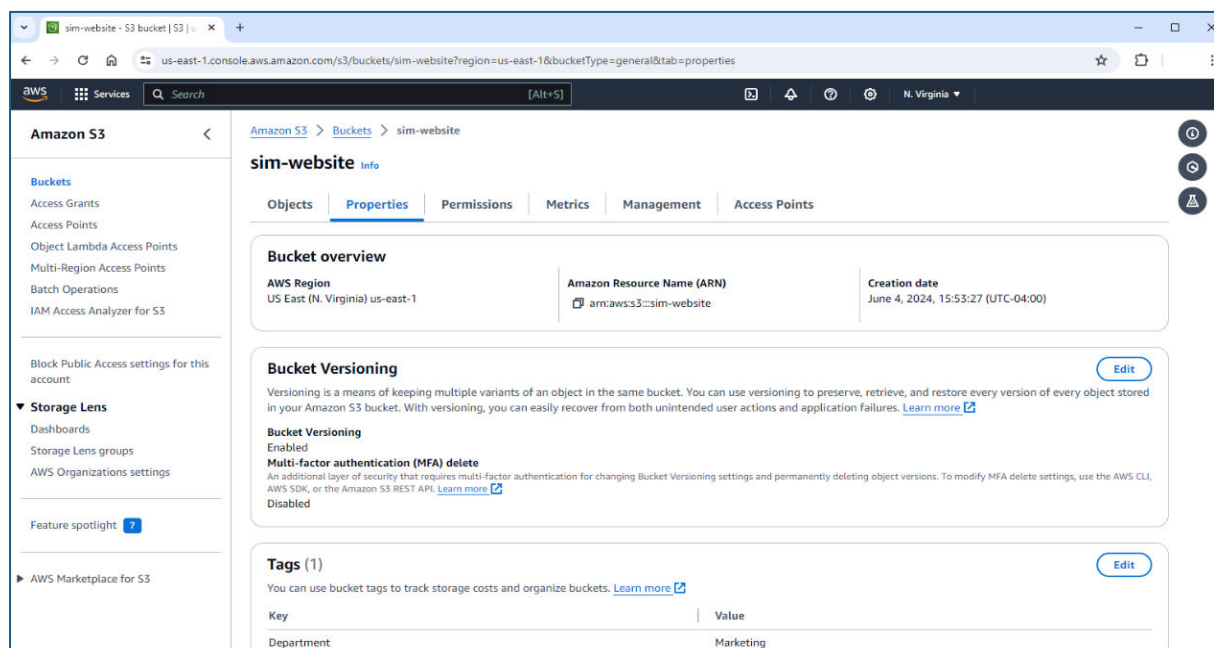
Index document: Enter index.html **Error document:** Enter error.html

18. Choose the scroll bar to scroll down, Choose Save changes.

19. Choose the scroll bar to scroll to the Static website hosting panel.

20. In the Static website hosting panel under Bucket website endpoint, choose the link. A new tab opens where you receive a 403 Forbidden message because you have not yet configured the bucket permissions.

21. Choose the AWS Management Console tab on your browser. You have configured your bucket to host a static website.



sim-website - S3 bucket | S3

us-east-1.console.aws.amazon.com/s3/buckets/sim-website?region=us-east-1&bucketType=general&tab=properties

Services

Search

[Alt+S]

N. Virginia

Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboards

Storage Lens groups

AWS Organizations settings

Feature spotlight: 7

AWS Marketplace for S3

Successfully edited static website hosting.

Amazon S3 > Buckets > sim-website

sim-website Info

Objects Properties Permissions Metrics Management Access Points

Bucket overview

AWS Region

US East (N. Virginia) us-east-1

Amazon Resource Name (ARN)

arn:aws:s3::sim-website

Creation date

June 4, 2024, 15:53:27 (UTC-04:00)

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

Enabled

Multi-factor authentication (MFA) delete

Disabled

Tags (1)

You can use bucket tags to track storage costs and organize buckets. [Learn more](#)

sim-website - S3 bucket | S3

us-east-1.console.aws.amazon.com/s3/buckets/sim-website?region=us-east-1&bucketType=general&tab=properties

Services

Search

[Alt+S]

N. Virginia

Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboards

Storage Lens groups

AWS Organizations settings

Feature spotlight: 7

AWS Marketplace for S3

Successfully edited static website hosting.

Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. [Learn more](#)

Object Lock

Disabled

Requester pays

When enabled, the requester pays for requests and data transfer costs, and anonymous access to this bucket is disabled. [Learn more](#)

Requester pays

Disabled

Static website hosting

Use this bucket to host a website or redirect requests. [Learn more](#)

Static website hosting

Enabled

Hosting type

Bucket hosting

Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://sim-website.s3-website-us-east-1.amazonaws.com>

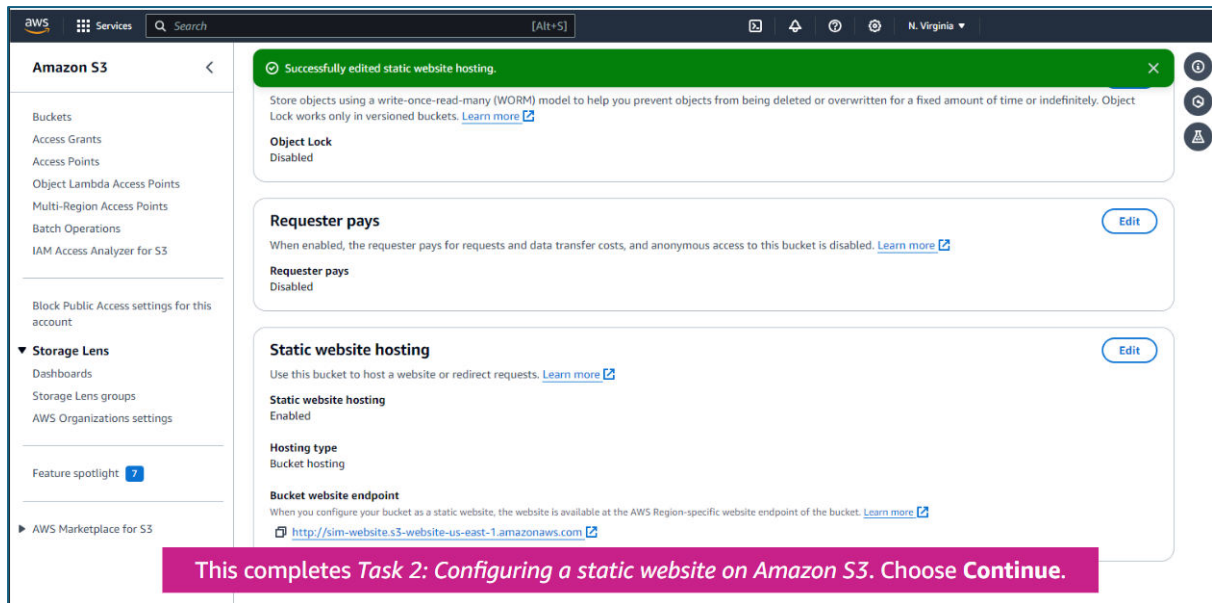
sim-website.s3-website-us-east-1.amazonaws.com

403 Forbidden

- Code: AccessDenied
- Message: Access Denied
- RequestId: 7FRGDT7RN2142KQT
- HostId: XjYpYgKAqE54Z3OZtao2+mTibzZQR6vEOA6F5vQT3zxOhr0PIW6qDzjo6iHRmnBdTs9lKCIWM=

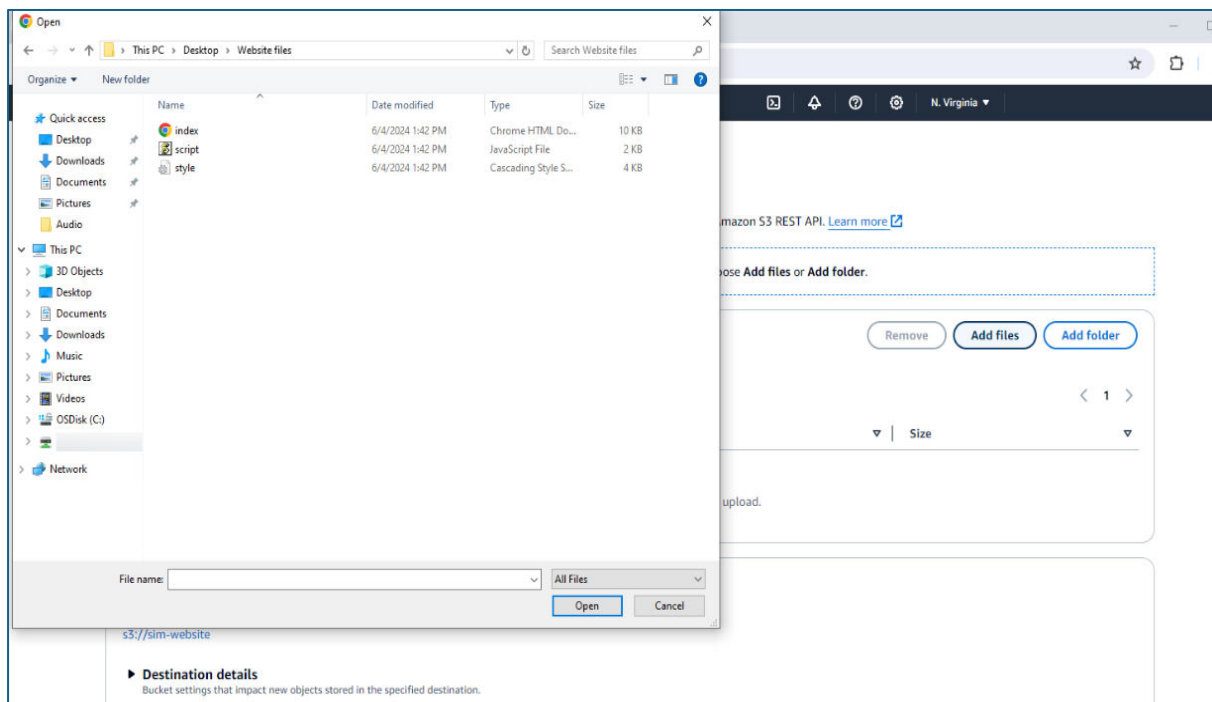
An Error Occurred While Attempting to Retrieve a Custom Error Document

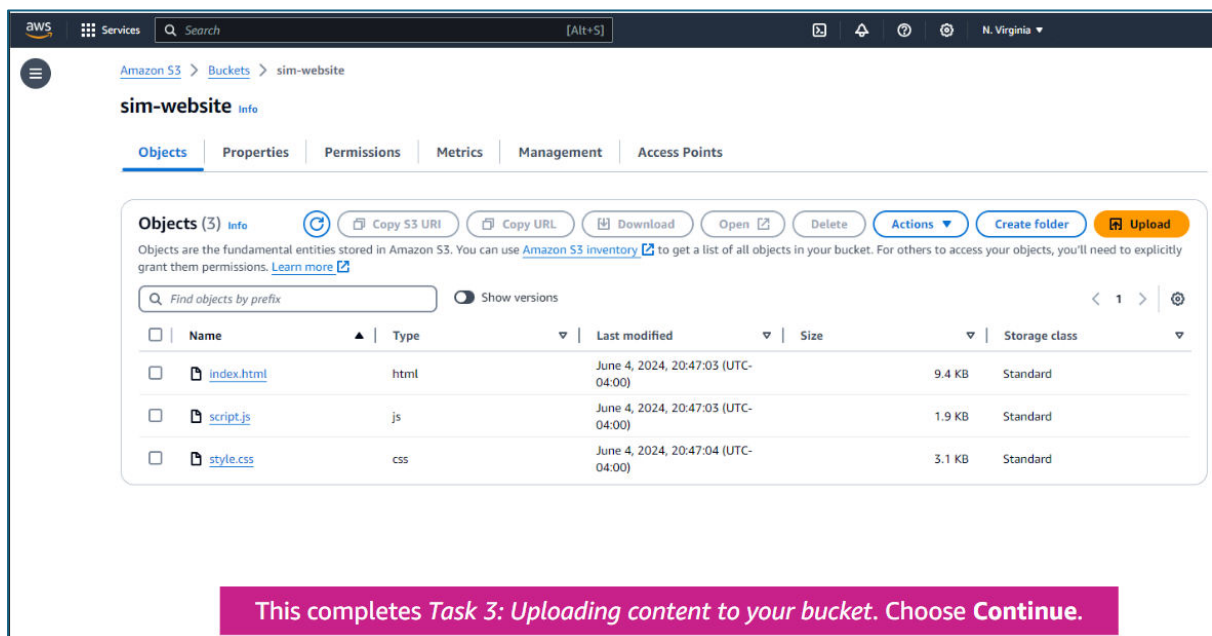
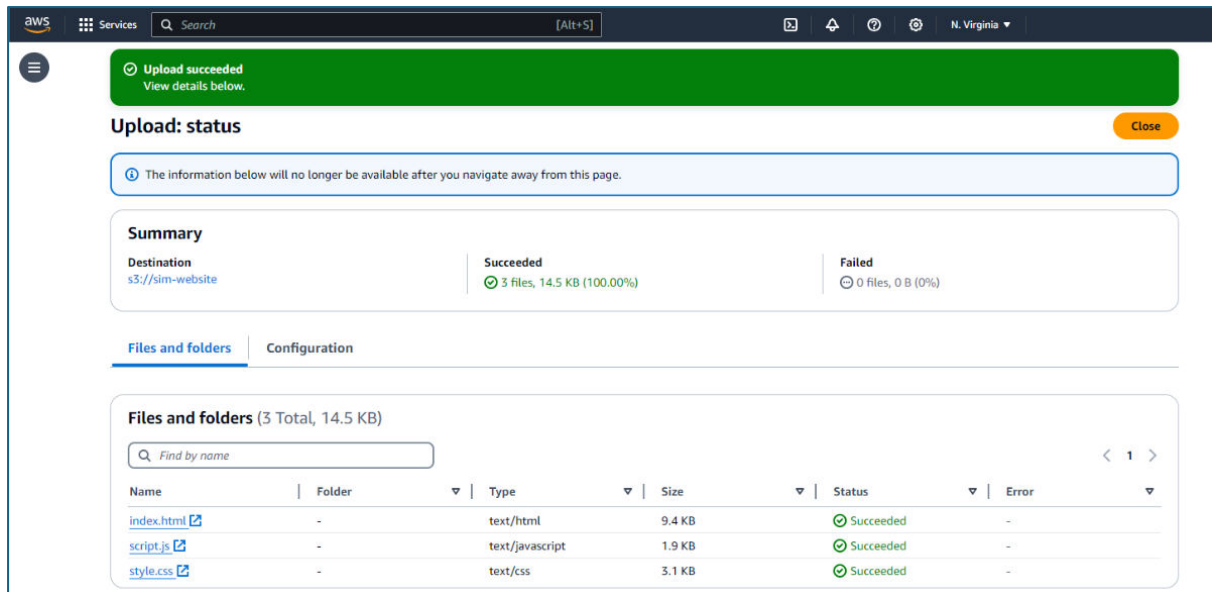
- Code: AccessDenied
- Message: Access Denied



Task 3: Uploading content to your bucket

22. Choose the scroll bar to scroll to the top of the page, and choose the Objects tab.
23. Choose Upload, Choose Add files, Choose the Website files folder, and choose Open to open the folder.
24. Use your mouse to choose each of the following files: index.html, script.js, and style.css (order does not matter). Then choose Open.
25. Choose the scroll bar to scroll down, Choose Upload. Your files are uploaded to the bucket, Choose Close.

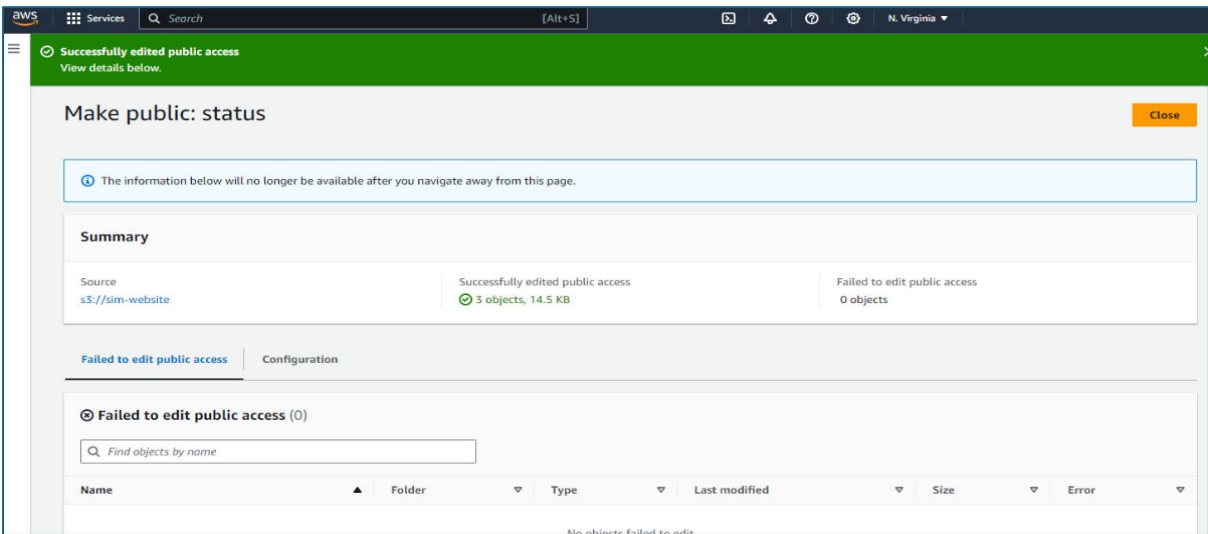
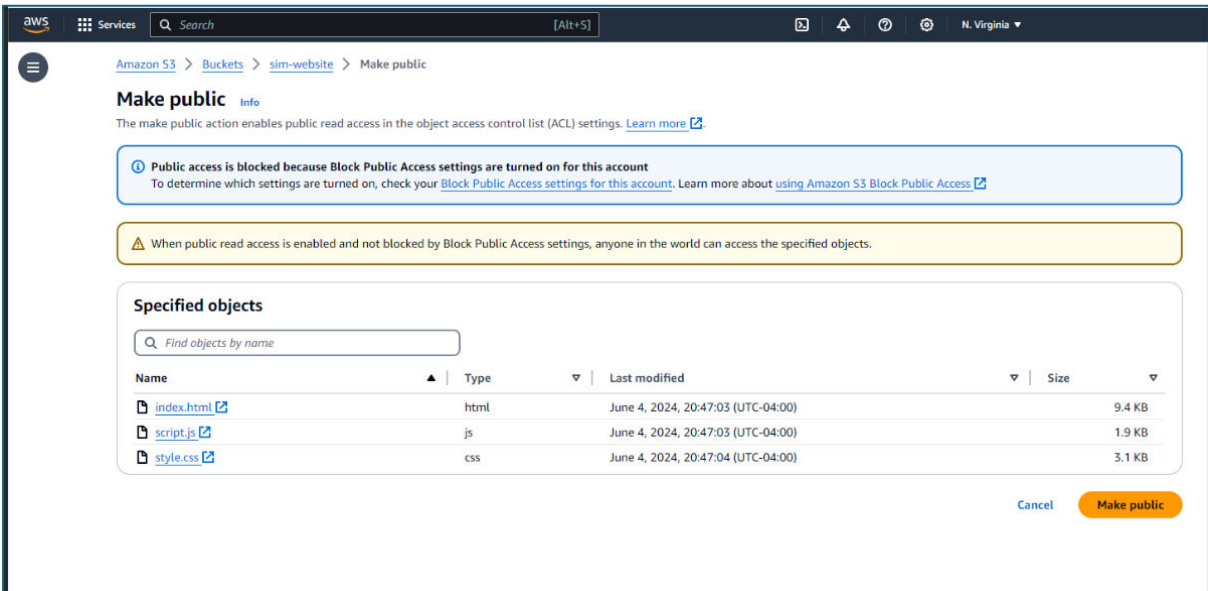
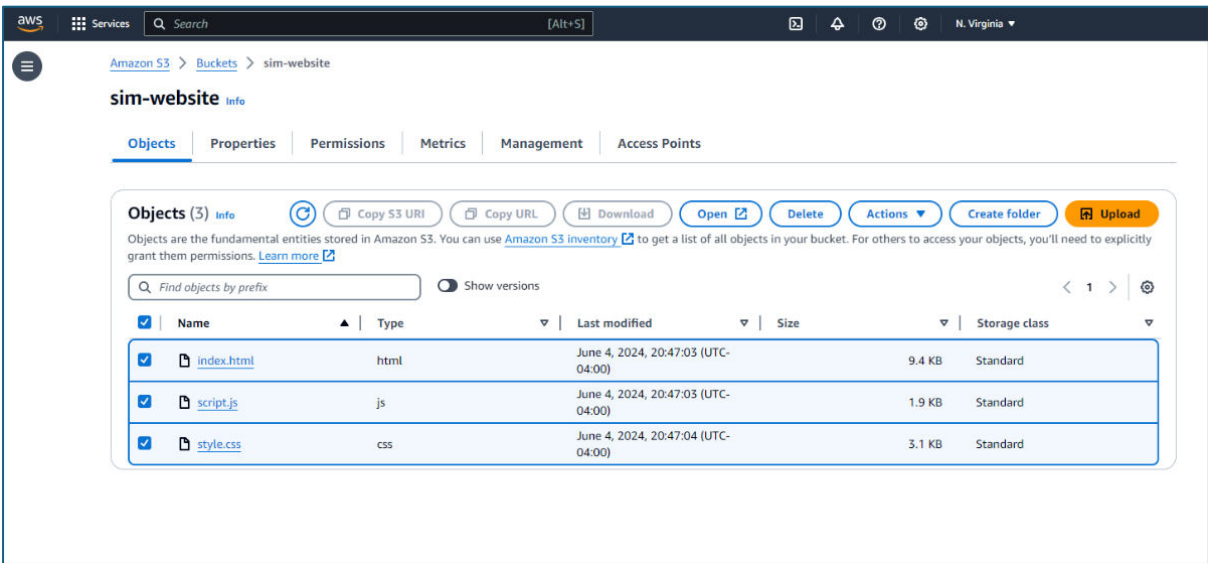


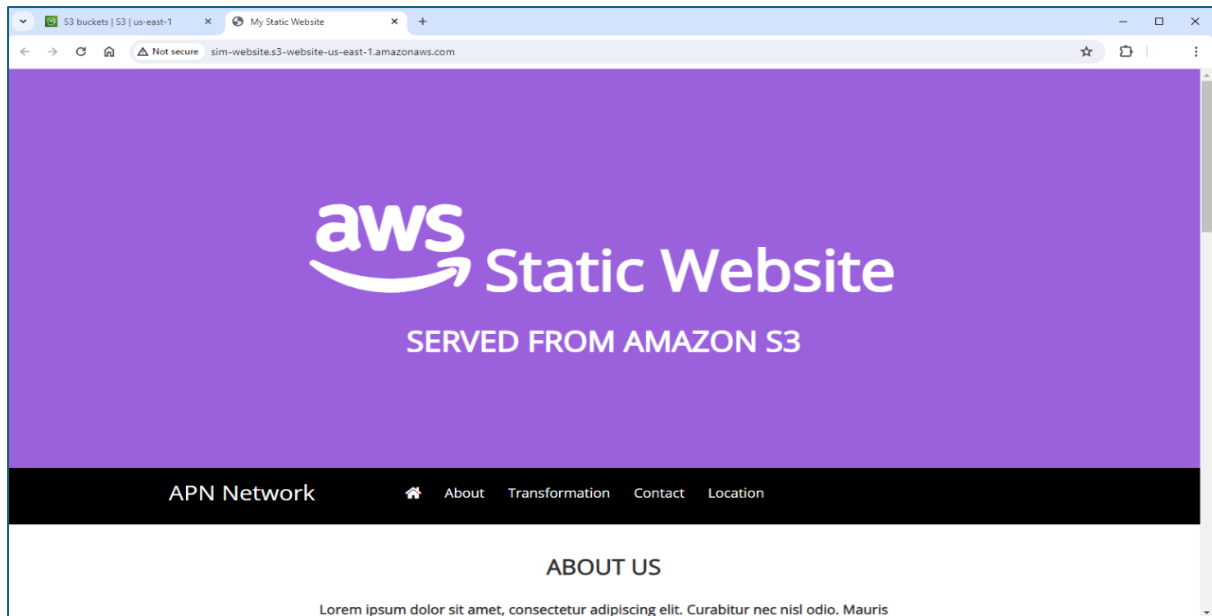


Task 4: Turning on public access to the objects

26. Return to the browser tab that showed the 403 Forbidden message.
27. Choose the Refresh button for the webpage. You should still see a 403 Forbidden message.
28. Keep the website tab open, and return to the web browser tab with the Amazon S3 console.
29. Choose the Name checkbox to select all three objects.
30. In the Actions menu, choose Make public using ACL. A list of the three objects is displayed.
31. Choose Make public. Your static website is now publicly accessible. Choose Close.

32.Return to the web browser tab that has the 403 Forbidden message; Refresh the webpage. You should now see the static website that Amazon S3 is hosting.





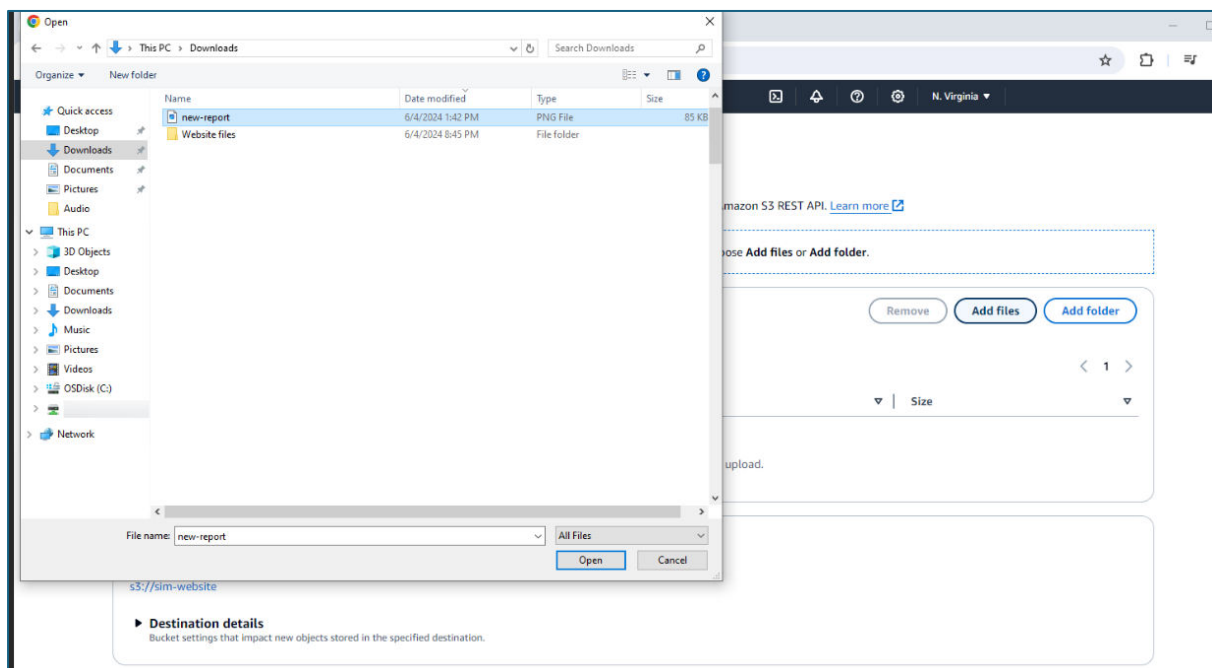
Task 5: Securely sharing an object by using a presigned URL

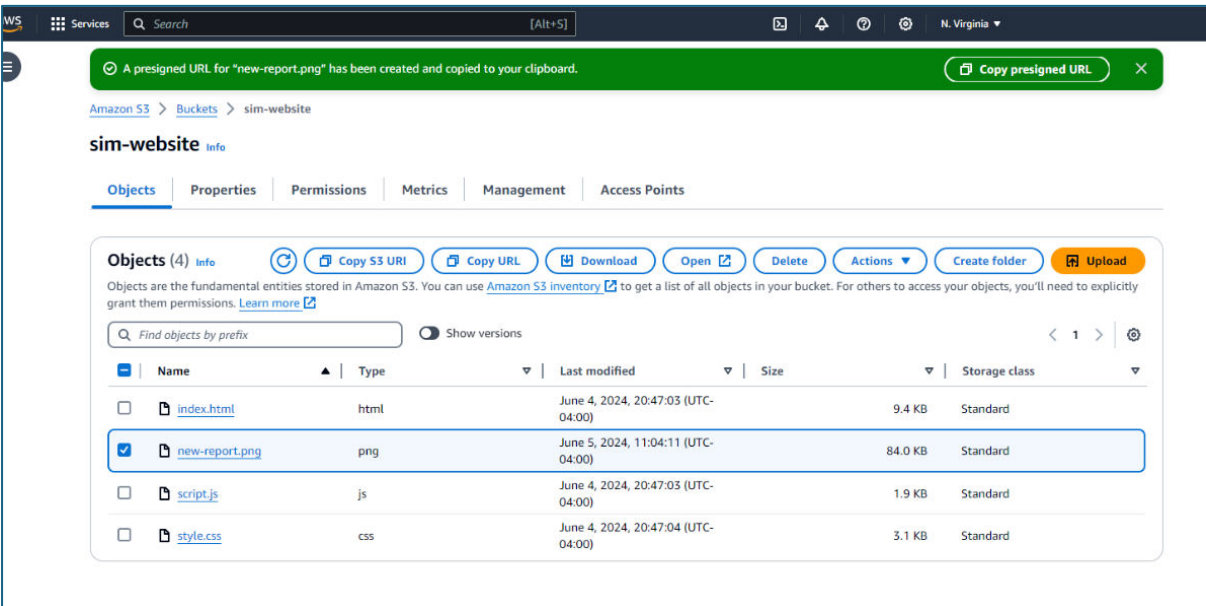
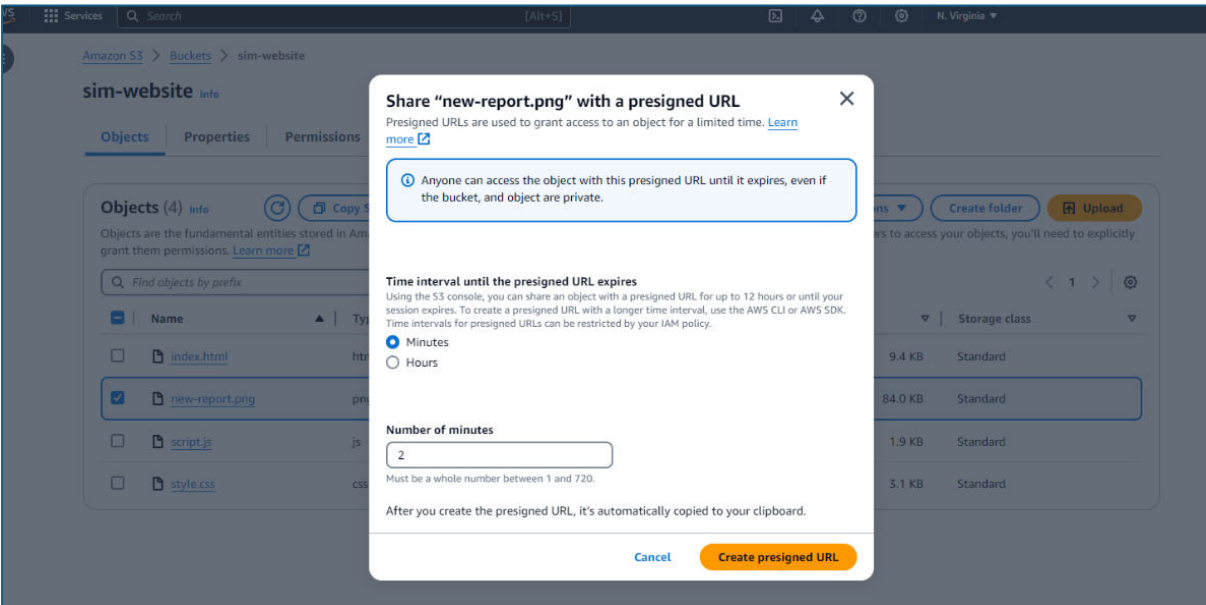
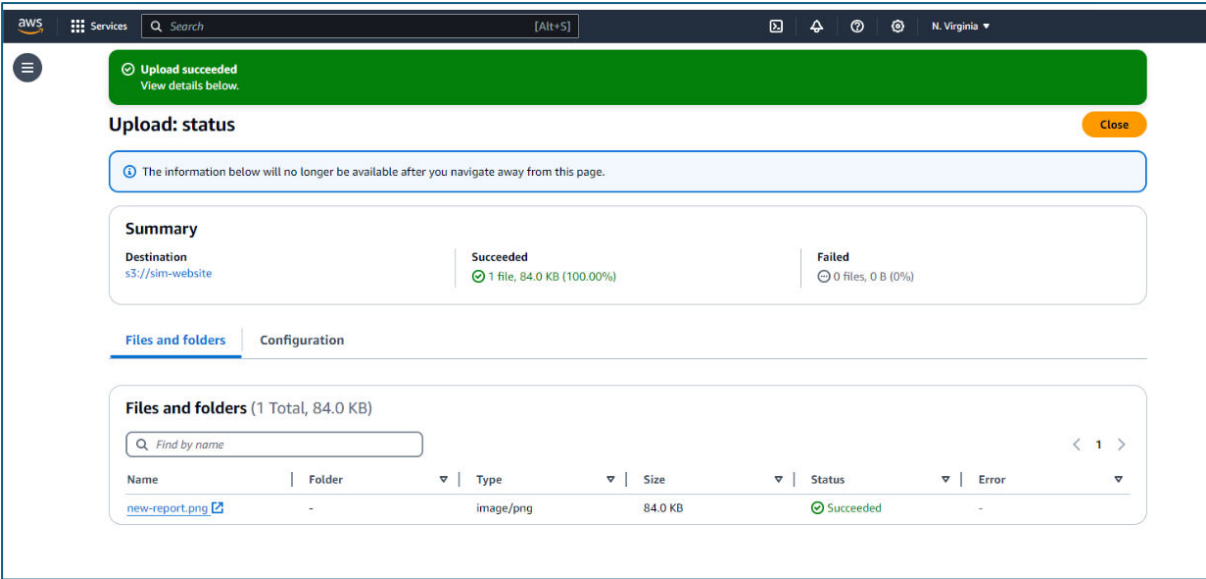
33. Choose Upload > Choose Add files > Choose the file new-report file and choose Open.

34. Choose the scroll bar to scroll down & Choose Upload then Choose Close.

35. In the Objects tab, choose new-report.png > From the Actions menu, select Share with a presigned URL > In the pop-up window, keep the default Minutes selected for the Time interval until the presigned URL expires. For Number of minutes, enter 2.

36. Choose Create presigned URL > From the banner at the top of the page, choose Copy presigned URL > Open a new browser tab > Paste the URL that you copied into the address bar.





Service	Operation	UsageType	Resource	StartTime	EndTime	UsageValue
AmazonS3	HeadBucket	USW2-C3DataTransfer-Out-Bytes	lab-test-bucket-77	10/31/2020 0:00	12/31/2020 11:59	15309
AmazonS3	PutObject	USW2-C3DataTransfer-In-Bytes	admin-test-77	10/31/2020 0:00	12/31/2020 11:59	19032
AmazonS3	HeadBucket	USW1-Requests-Tier2	admin-test-77	10/31/2020 0:00	12/31/2020 11:59	128
AmazonS3	PutObjectForReplication	USW1-Requests-SIA-Tier1	mybucket-98765	10/31/2020 0:00	12/31/2020 11:59	56888
AmazonS3	GetObjectForReplication	USW2-USW2-AWS-In-Bytes	mybucket-98766	10/31/2020 0:00	12/31/2020 11:59	254587
AmazonS3	HeadBucket	USW2-C3DataTransfer-Out-Bytes	mybucket-98767	10/31/2020 0:00	12/31/2020 11:59	235
AmazonS3	PutObject	USW2-C3DataTransfer-In-Bytes	mybucket-98768	10/31/2020 0:00	12/31/2020 11:59	25589
AmazonS3	PutObject	USW2-Requests-Tier2	mybucket-98769	10/31/2020 0:00	12/31/2020 11:59	2348
AmazonS3	PutObjectForReplication	USW1-Request-SIA-Tier1	mybucket-98770	10/31/2020 0:00	12/31/2020 11:59	15309
AmazonS3	GetObjectForReplication	USW1-USW2-AWS-In-Bytes	mybucket-98771	10/31/2020 0:00	12/31/2020 11:59	19032
AmazonS3	GetObjectForReplication	USW2-C3DataTransfer-Out-Bytes	lab-example-bucket	10/31/2020 0:00	12/31/2020 11:59	128
AmazonS3	HeadBucket	USW2-C3DataTransfer-In-Bytes	lab-example-bucket	10/31/2020 0:00	12/31/2020 11:59	56888
AmazonS3	PutObject	USW2-Requests-Tier2	lab-example-bucket	10/31/2020 0:00	12/31/2020 11:59	254587
AmazonS3	PutObjectForReplication	USW1-Request-SIA-Tier1	lab-example-bucket	10/31/2020 0:00	12/31/2020 11:59	235
AmazonS3	GetObjectForReplication	USW1-USW2-AWS-In-Bytes	lab-example-bucket	10/31/2020 0:00	12/31/2020 11:59	25589

A presigned URL for "new-report.png" has been created and copied to your clipboard. [Copy presigned URL](#)

[Presigned URL copied](#)

Amazon S3 > Buckets > sim-website

sim-website info

[Objects](#) [Properties](#) [Permissions](#) [Metrics](#) [Management](#) [Access Points](#)

Objects (4) info [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix ☐ Show versions

Name	Type	Last modified	Size	Storage class
☐ index.html	html	June 4, 2024, 20:47:03 (UTC-04:00)	9.4 KB	Standard
☒ new-report.png	png	June 5, 2024, 11:04:11 (UTC-04:00)	84.0 KB	Standard
☐ script.js	js	June 4, 2024, 20:47:03 (UTC-04:00)	1.9 KB	Standard
☐ style.css	css	June 4, 2024, 20:47:04 (UTC-04:00)	3.1 KB	Standard

This completes Task 5: Securely sharing an object by using a presigned URL. Choose Continue.

Task 6: Using a bucket policy to secure your bucket

37. Choose the Permissions tab > Choose the scroll bar to scroll down to the Bucket policy panel > In the Bucket policy panel, choose Edit.

38. Copy the following policy text and paste it in the Policy text editor field.

39. Choose the scroll bar to scroll down > Choose Save changes > Return to the Objects tab > Select index.html > Choose Delete > In the Delete objects panel, enter delete to confirm that you want to remove this file.

40. Choose Delete objects. Notice that the index.html file is listed in the Failed to delete pane. This entry confirms that your policy is working and preventing the website's files from being deleted.

41. Choose Close to return to the Objects tab. Your bucket policy is now protecting your website files from being deleted.

aws

Services

Search

[Alt+S]

N. Virginia

Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboards

Storage Lens groups

AWS Organizations settings

Feature spotlight

AWS Marketplace for S3

Amazon S3 > Buckets > sim-website > Edit bucket policy

Info

Edit bucket policy

Bucket policy

Policy examples

Policy generator

The bucket policy, written in JSON, provides access to the objects stored in the bucket. Bucket policies don't apply to objects owned by other accounts. [Learn more](#)

Bucket ARN

arn:aws:s3::sim-website

Policy

Public access is blocked because Block Public Access settings are turned on for this account

To determine which settings are turned on, check your [Block Public Access settings for this account](#). [Learn more about using Amazon S3 Block Public Access](#)

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

"Version": "2012-10-17",

"Id": "MyBucketPolicy",

"Statement": [

{

"Sid": "BucketPutDelete",

"Effect": "Deny",

"Principal": "*",

"Action": "s3:DeleteObject",

"Resource": [

"arn:aws:s3::sim-website/index.html",

"arn:aws:s3::sim-website/script.js",

"arn:aws:s3::sim-website/style.css"

Edit statement

Select a statement

Select an existing statement in the policy or add a new statement.

+ Add new statement

aws

Services

Search

[Alt+S]

N. Virginia

Amazon S3

Buckets

Access Grants

Access Points

Object Lambda Access Points

Multi-Region Access Points

Batch Operations

IAM Access Analyzer for S3

Block Public Access settings for this account

Storage Lens

Dashboards

Storage Lens groups

AWS Organizations settings

Feature spotlight

AWS Marketplace for S3

sim-website

Info

sim-website

Objects | Properties | Permissions | Metrics | Management | Access Points

Permissions overview

Access finding

Access findings are provided by IAM external access analyzers. [Learn more about How IAM analyzer findings work](#)

[View analyzer for us-east-1](#)

Block public access (bucket settings)

Edit

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to all your S3 buckets and objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to your buckets or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

Block Public Access settings for this account are currently turned on

[Block Public Access settings for this account](#) that are enabled apply even if they are disabled for this bucket.

Block all public access

Off

Individual Block Public Access settings for this bucket

aws

Services

Search

[Alt+S]

N. Virginia

Amazon S3 > Buckets > sim-website > Delete objects

Info

Delete objects

If a folder is selected for deletion, all objects in the folder will be deleted, and any new objects added while the delete action is in progress might also be deleted. If an object is selected for deletion, any new objects with the same name that are uploaded before the delete action is completed will also be deleted. [Learn more](#)

Deleting the specified objects adds delete markers to them

If you need to undo the delete action, you can delete the delete markers. [Learn more](#)

Specified objects

Find objects by name

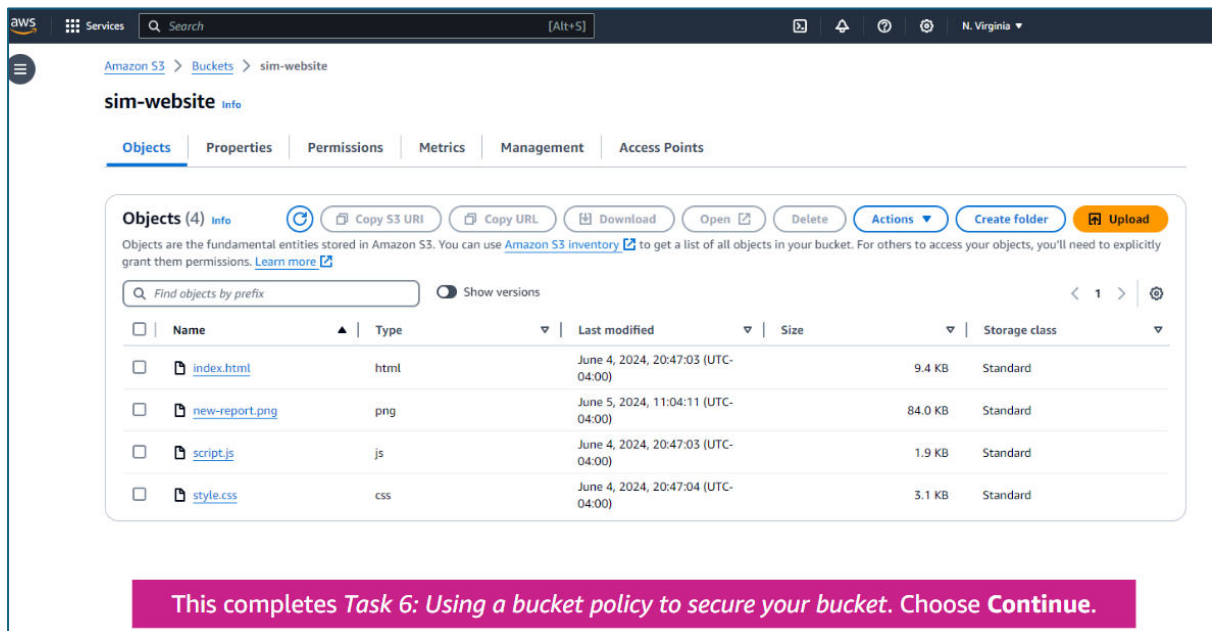
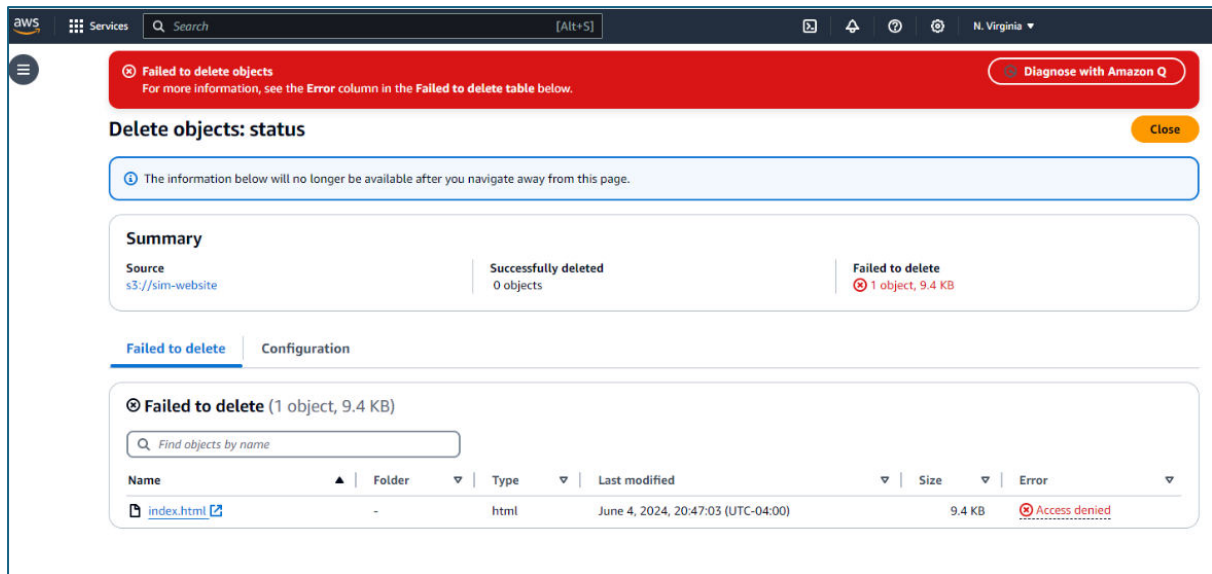
Name	Type	Last modified	Size
index.html	html	June 4, 2024, 20:47:03 (UTC-04:00)	9.4 KB

Delete objects?

To confirm deletion, type delete in the text input field.

delete

Cancel Delete objects



Task 7: Updating the website

42. On your computer, load the index.html file into a text editor (in this simulation, you use Notepad).

43. Find the text Served from Amazon S3, and replace it with Created by Jane.

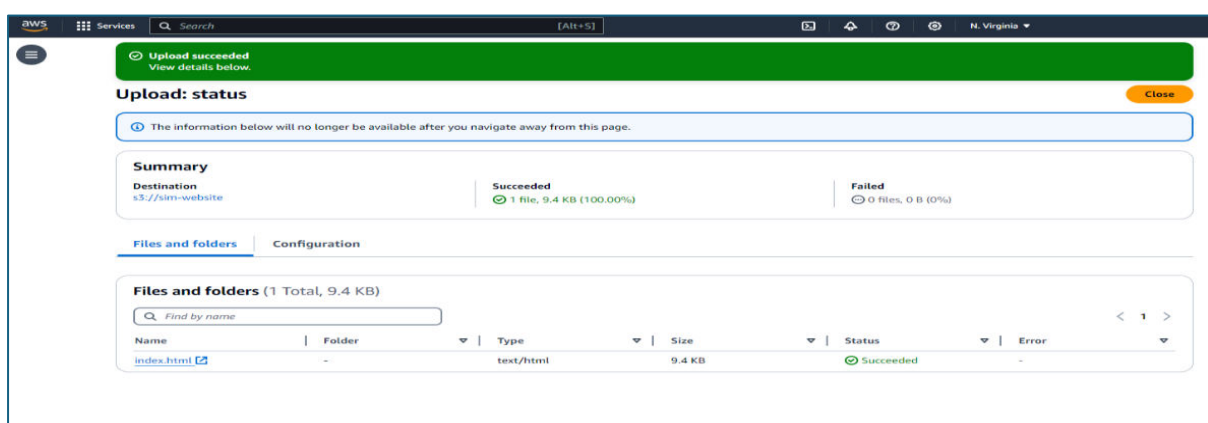
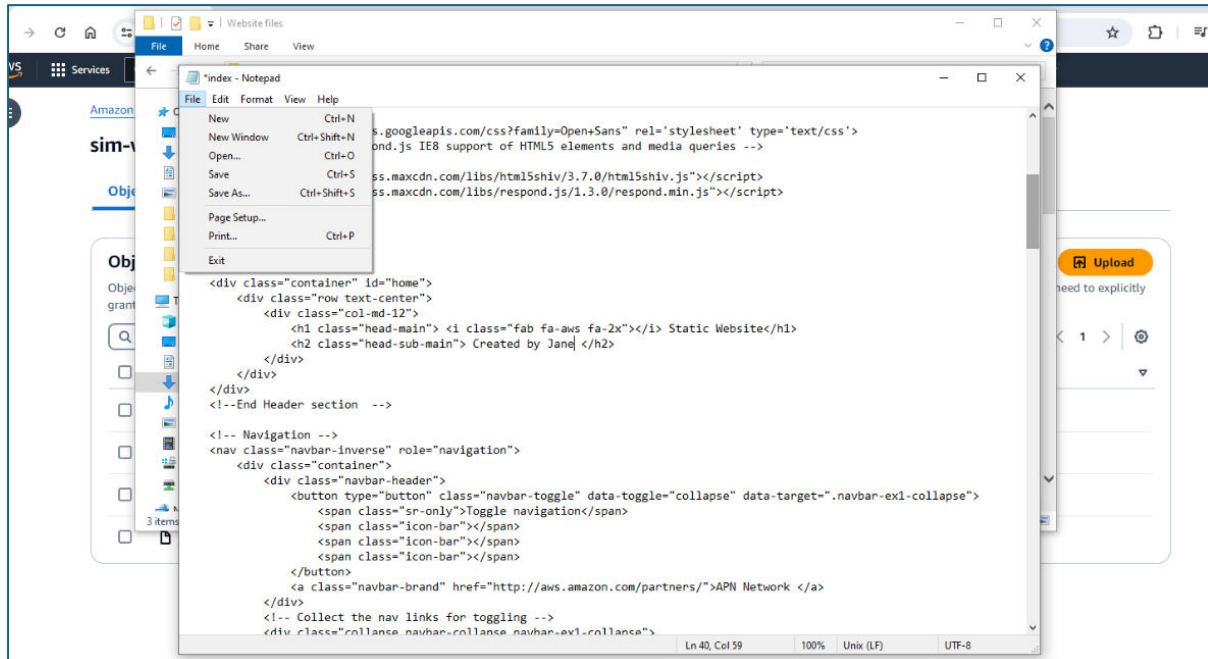
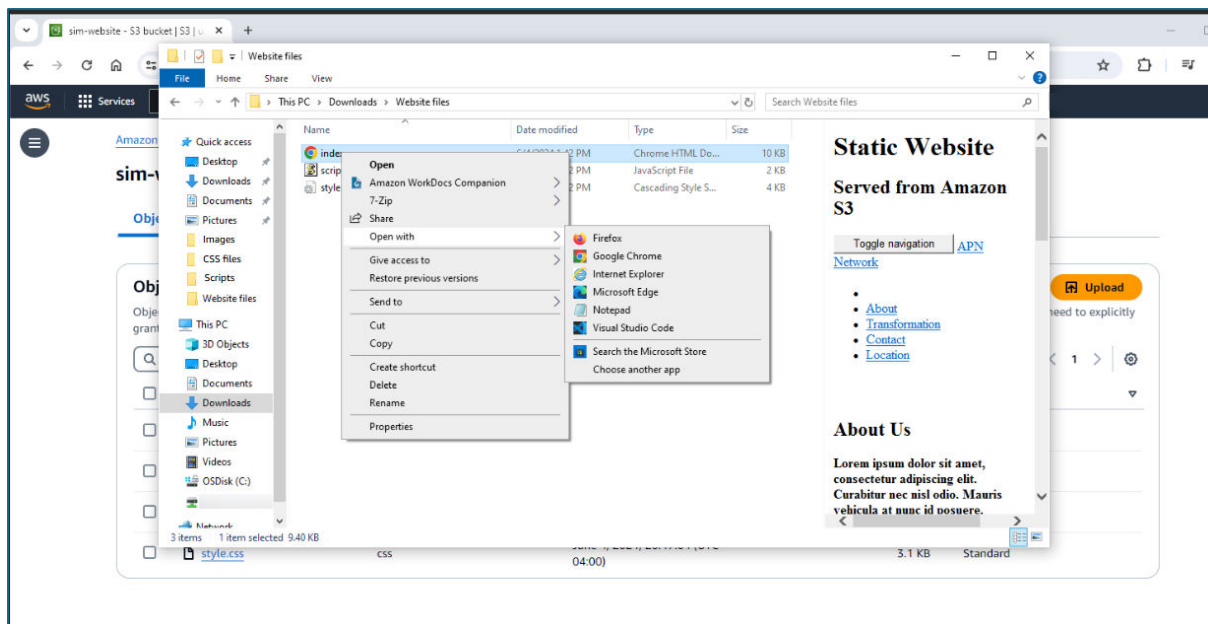
44. Choose the index.html file name (choose the link, not the checkbox).

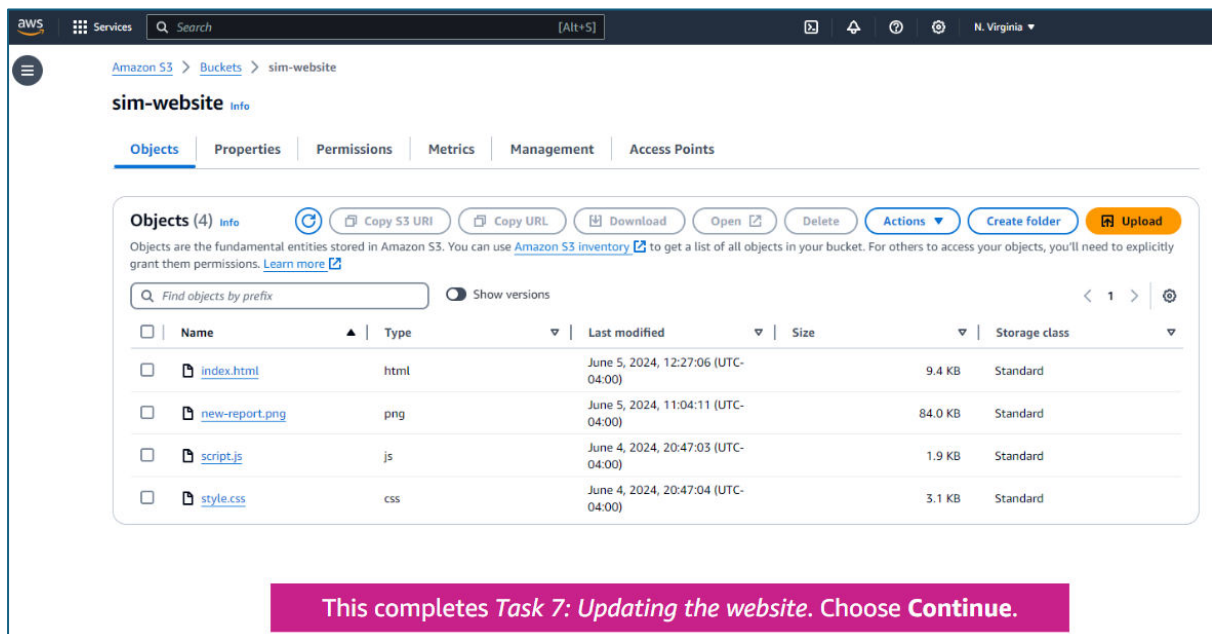
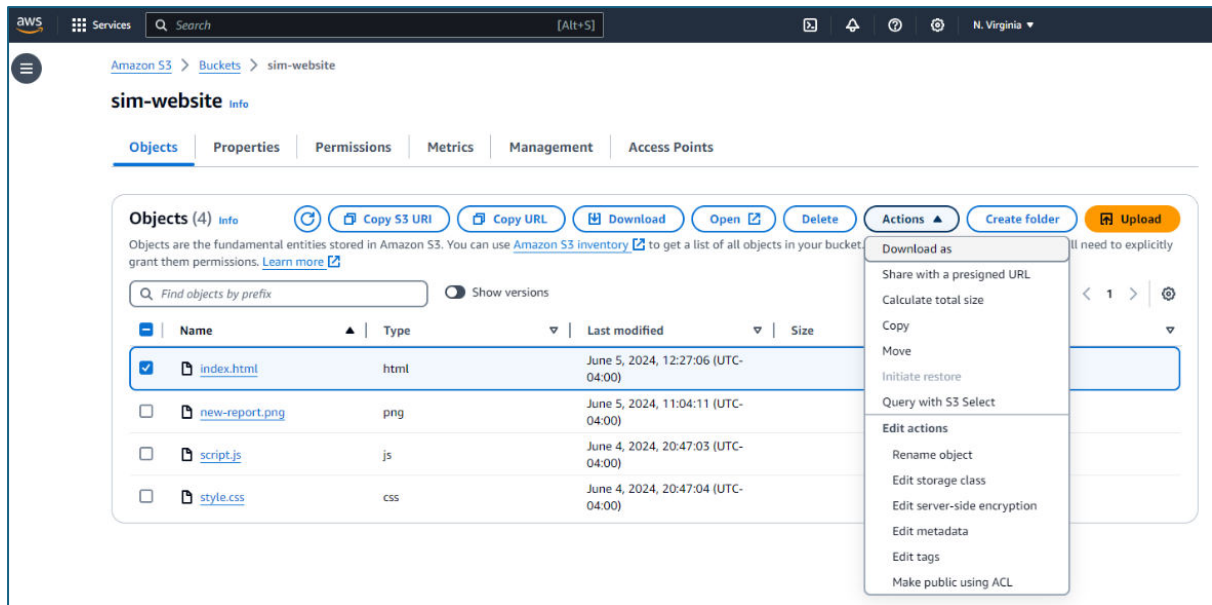
45. Choose the Back arrow on your browser to return to the Amazon S3 console. 54. Choose the sim-website link from the navigation at the top of the page.

46. Upload the index.html file that you just edited.

47. Select the index.html checkbox, and in the Actions menu, choose the Make public using ACL option again.

48. Choose Make public, and choose Close.

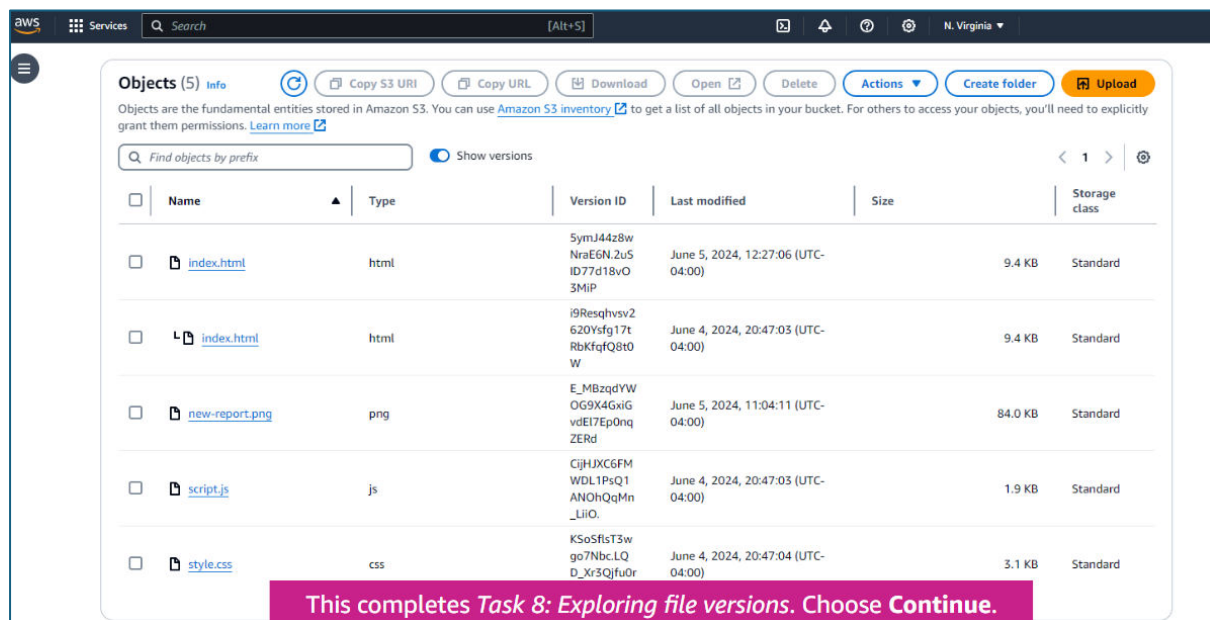
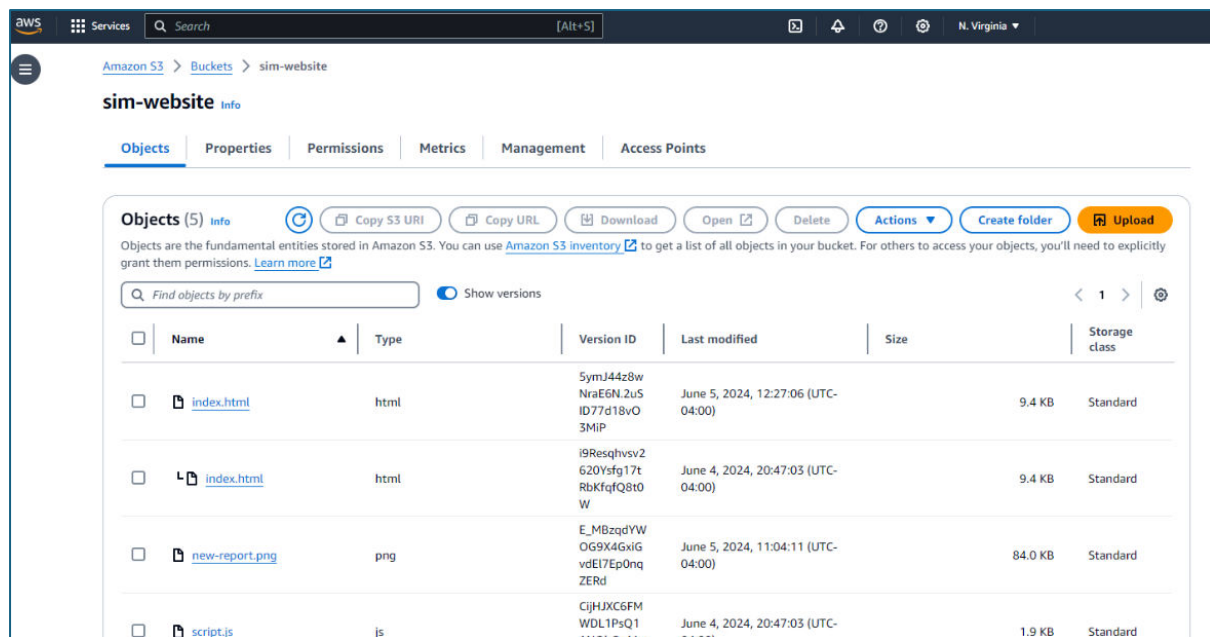




Task 8: Exploring file versions

49. Choose Show versions to see which files have multiple versions > Choose the scroll bar to scroll down.

50. Review the list of objects in the bucket. Notice that each file has a Version ID.



Conclusion :

Through this activity, we successfully created and configured an Amazon S3 bucket for static website hosting. We uploaded website files, enabled public access, and used bucket policies to secure the content. Additionally, we explored features like presigned URLs and versioning for better content management. This exercise demonstrated the scalability, flexibility, and security capabilities of Amazon S3 for hosting and managing web content in the cloud.